



# Training attentive individuation leads to visuo-spatial working memory improvement in low-performing older adults: An online study

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## Abstract

Cognitive decrements are typical of physiological aging. Among these age-related cognitive changes, visuo-spatial working memory (vWM) decline has a prominent role due to its effects on other cognitive functions and daily routines. To reinforce vWM in the aging population, several cognitive training interventions have been developed in the past years. Given that vWM functioning depends (at least partially) on the efficiency of attention selection of the relevant objects, in the present study we implemented a short (five sessions), online intervention that primarily trained attentive individuation of target items and tested training effects on a vWM task. Attention training effects were compared with practice (i.e., a group that repeatedly performed the same vWM task) and test–retest effects (i.e., a passive group). After the training, the results showed attention training effects of the same magnitude as practice effects, confirming that the enhancement of attentive individuation has a positive cascade influence on maintaining items in vWM. Moreover, training and practice effects were only evident in low-performing older adults. Thus, interindividual differences at baseline crucially contribute to training outcomes and are a fundamental factor to be accounted for in the implementation of cognitive training protocols.

**Keywords** Working memory · Cognitive aging · Cognitive training · Attentive individuation · Interindividual differences

Working memory (WM) capacity, namely the ability to retain and simultaneously manipulate the retained material (Baddeley, 2010), typically decreases in physiological aging (Salthouse, 2010). Moreover, visuo-spatial WM (vWM) is typically more affected than verbal WM by increasing age (Jenkins et al., 2000). Given that WM largely contributes to the functioning of other aspects of human cognition (e.g., Baudouin et al., 2009; Borella et al., 2011; Chen & Li, 2007; Park et al., 2002; Unsworth et al., 2005) and has a prominent role in everyday life (Hahn & Lachman, 2015), in recent years, researchers have actively investigated the possibility of improving WM capacity through cognitive training interventions in the aging population, although with sometimes contradictory results on their efficacy (see, for reviews, Karbach & Verhaeghen, 2014; Melby-Lervåg & Hulme, 2016; Teixeira-Santos et al., 2019).

Several studies (Awh et al., 2006; Gazzaley & Nobre, 2012; Olivers, 2008) highlighted a tight link between vWM and visual attention. Indeed, the activation of the two processes broadly overlaps at the neural level, and results in a right frontoparietal network (Awh & Jonides, 2001; LaBar et al., 1999). Moreover, several neuroimaging studies (Rutman et al., 2010; Vogel et al., 2005; Zanto & Gazzaley, 2009) showed that attention modulation correlates with vWM, so that those individuals who are more efficient in selecting relevant items exhibit a larger vWM capacity. Finally, a key feature of both vWM and visual attention is a capacity limit at approximately four/five items (Cavanagh & Alvarez, 2005; Cowan, 2010). Taken together, these pieces of evidence led to the proposal that attention (at least, partly) contributes to the amount (and quality) of information that we encode in the vWM buffer (see Fougny & Marois, 2006).

Importantly, the link between attention and vWM has also been shown in the older population. Indeed, accumulating results point to the fact that age-related deficits in attention are associated with the vWM impairment observed in older adults (Gazzaley et al., 2005; Gazzaley et al., 2008; Störmer et al., 2013; Tagliabue et al., 2019).

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Given these premises, within a cognitive training perspective reinforcing attention might thus represent a fruitful approach to enhance vWM in the normal aging population. In line with this hypothesis, in a previous electrophysiological (EEG) study (Tagliabue et al., 2020), we exploited event-related potentials (ERPs) to pinpoint which cognitive processes were modulated over a 4-days practice period of a delayed match-to-sample (DMTS) judgment. The DMTS is a task typically used to estimate the maximum number of items one can efficiently maintain in the memory buffer (Vogel & Machizawa, 2004). To perform it, individuals must serially deploy various cognitive processes: the initial stimulus encoding, an attention-based mechanism for target individuation, and, finally, active memory retention. The EEG data indicated that the practice-related increase in vWM capacity observed in older adults was accompanied by neural modifications not only during item retention in the memory buffer (reflected by modulations of the CDA component; Vogel & Machizawa, 2004), but also in the previous processing stage of relevant item individuation (reflected by modulations of the N2pc component; Eimer, 1996). These results suggest that selectively boosting the attentional ability to individuate target memoranda could have a beneficial cascade effect on vWM capacity.

However, despite the large amount of evidence linking attention and vWM, only a few intervention studies (Arend & Zimmer, 2012; Schmicker et al., 2016; Schmicker et al., 2021; Shin et al., 2015; Thompson et al., 2013) investigated whether training attention mechanisms could improve vWM, even when individuals do not receive explicit requirements for object memorization. Indeed, findings are mixed and mostly limited to younger adulthood.

To test the hypothesis that training attention may lead to improvements in vWM, in the current study we implemented a short (five sessions; see Schmicker et al., 2016) online training using enumeration, a task that primarily engages attentive individuation. In enumeration tasks, individuals are very rapid and/or accurate with small sets of items (up to three/four), a phenomenon called ‘subitizing’ (Kaufman et al., 1949). Beyond that range, enumeration becomes more effortful with each additional item (‘counting’; Mandler & Shebo, 1982). Notably, attentional constraints play a key role in enumeration and in the generation of the subitizing phenomenon: the individual subitizing range appears to be determined by the number of objects one can simultaneously attend to (e.g., Ester et al., 2012; Pagano et al., 2014)—that is, the higher this number, the larger the subitizing range. Moreover, several pieces of evidence confirmed that enumeration requires attention to occur (Chen et al., 2022; Olivers & Watson, 2008; Railo et al., 2008). Various neuroimaging findings further resonate with the involvement of attention in rapid enumeration. EEG data (e.g., Ester et al., 2012; Pagano et al., 2014; Pagano & Mazza, 2012; Wurm et al., 2021; see Mazza &

Caramazza, 2015, for a review) have shown the contribution of an attention-based individuation mechanism, as indexed by the N2pc component and its modulation, to the occurrence of subitizing. Similarly, enumeration in the subitizing range activates the temporoparietal junction, an area known to be implicated in stimulus-driven attention (Ansari et al., 2007; Vetter et al., 2011).

In the present study, older participants were trained with an adaptive enumeration task to enlarge their subitizing range by taxing the attentive individuation mechanism that typically precedes item retention in the DMTS task. Enumeration training benefits were then directly tested on the DMTS task (Vogel & Machizawa, 2004), as a measure of vWM capacity. Two control groups were also included, one engaged in an adaptive DMTS task (i.e., practice effect) and a passive control group (i.e., test–retest effect).

In training interventions, individual differences in cognitive abilities at baseline (i.e., the pretraining cognitive level) are a critical factor, as they represent a crucial source of variability in determining training outcomes (see Ophey et al., 2020; Shaw & Hosseini, 2020; Traut et al., 2021, for recent reviews). However, the relationship between baseline performance and training gain is not unidirectional. Indeed, there is evidence supporting both the so-called magnification and compensation views (Lövdén et al., 2012). The magnification view states that high-performing individuals will experience greater benefit from training, as they can count on more cognitive resources to implement effortful strategies. On the other hand, the compensation perspective predicts that low performers are those showing larger training gains, since they have more room for improvement. This discrepancy seems to be solved when considering qualitative differences in cognitive regimens: magnification effects are more often evident in episodic memory training with a strategy-based (i.e., teaching an explicit strategy) approach (e.g., Brehmer et al., 2007; Verhaeghen et al., 1992); compensation effects usually emerge in executive function (including WM) trainings that primarily rely on a process-based (i.e., training a specific cognitive process) approach (e.g., Borella et al., 2017; Zinke et al., 2014). For all these reasons, in the present study, we evaluated whether individual differences at baseline may influence training outcomes by further dividing each participant sample into low and high performers.

In sum, in the present study, we aimed to test whether primarily training attentive individuation (through an enumeration task) is sufficient to improve vWM (i.e., attention training effect) of older adults, even when no explicit requirement to memorize the objects is provided. Finally, in line with previous literature (Ophey et al., 2020; Shaw & Hosseini, 2020; Traut et al., 2021), given the process-based nature of the training we expected low-performing older individuals to benefit more from training than high performers.

## Materials and methods

### Participants

The study was conducted online and participants were enrolled through the recruitment platform Prolific (<https://www.prolific.co/>). A sample size of 36 participants (i.e., 18 per performance level) per training group was estimated on the basis of a power analysis computed with PANGAEA (<https://jakewestfall.shinyapps.io/pangea/>): 80% power (Cohen, 1988) for a three-way interaction (Session  $\times$  Training  $\times$  Performance level; see Statistical Analyses for details) with a medium effect size (Cohen's  $d = .54$ ). The data of a subsample of participants (i.e., Passive and DMTS group) were analyzed through different measures in an already published study (Tagliabue et al., 2022) aimed at investigating age-related differences in online and offline learning.

Participants self-declared that they did not suffer from neurological or mental illnesses and that they had normal or corrected-to-normal vision. Additionally, they were screened with a custom-made online version of SATURN (Self-Administered Tasks Uncovering Risk of Neurodegeneration; Bissig et al., 2020) to exclude any potential cognitive impairment. Only those with a score  $\geq 26$  in SATURN were enrolled. A total of 130 participants initially participated in the study and were randomly assigned to one of the three groups. However, 23 of them dropped out or had technical issues and two did not complete the sessions in due time. Moreover, one participant was identified as an outlier ( $\pm 3$  SD) based on his performance during the first session (Kable et al., 2017). The final sample thus comprised 104 participants: 36 in the Passive group (mean age  $\pm$  SD =  $68.25 \pm 2.73$ , 15 females), 35 in the Enumeration group (mean age  $\pm$  SD =  $68.77 \pm 3.40$ , 18 female) and 33 in the DMTS group (mean age  $\pm$  SD =  $68.12 \pm 3.29$ , 17 females). The three groups did not differ in age ( $p = .637$ ), education ( $p = .894$ ) and SATURN score ( $p = .750$ ).

The study was approved by the Ethics Committee of the University of Trento and conducted in accordance with the 2013 Declaration of Helsinki. Participants' informed consent was collected prior to testing through the platform PsyToolkit (Stoet, 2010, 2017).

### Training structure

The training lasted nine days. During Sessions 1 and 7, all participants performed the DMTS task. During the intermediate sessions (from 2 to 6), all groups (including the Passive one) filled in some questionnaires. The Passive group was thus engaged in a daily activity as the other two groups. The DMTS and the Enumeration groups additionally performed an adaptive version of a DMTS and Enumeration task, respectively (see Stimuli and Procedure). In between Sessions 6 and

7 there was a 2-days break. Approximately 3 months after the execution of Session 7 (Brehmer et al., 2012; Li et al., 2008; Mahncke et al., 2006), participants were invited to a follow-up session and asked to perform the same DMTS task performed during Sessions 1 and 7 (Fig. 1a).

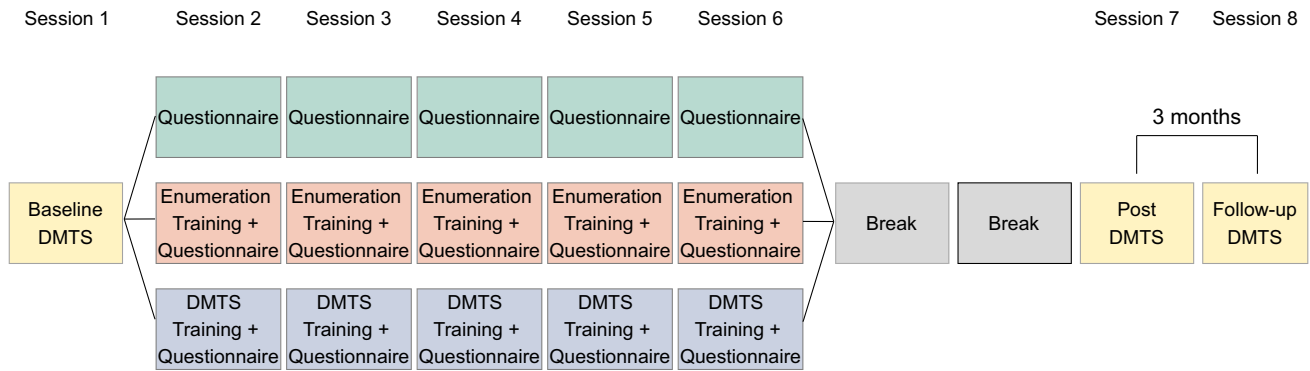
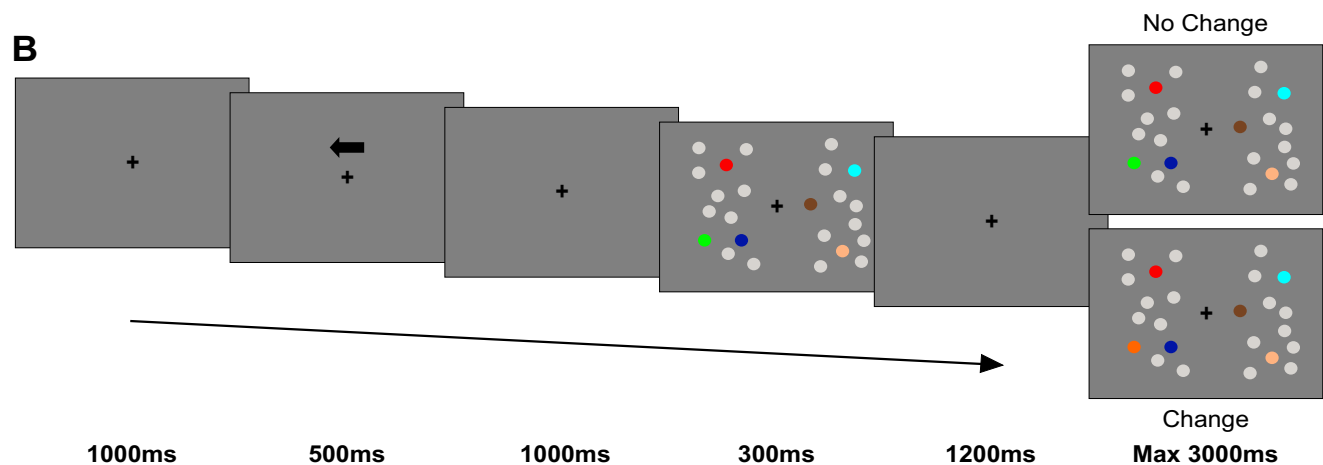
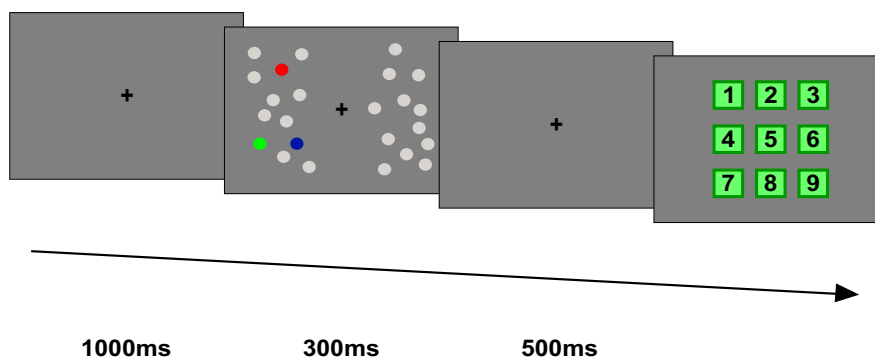
### Stimuli and procedure

All tasks were coded with PsychoPy (Peirce et al., 2019) and uploaded on Pavlovia (<https://pavlovia.org/>). Pavlovia has proved to be one of the most precise online experiment platforms in terms of stimulus duration and collection of reaction times, across different browsers and operating systems (Anwyl-Irvine et al., 2021; Bridges et al., 2020). At the beginning of each session, participants were required to sit in front of the monitor at arm length distance ( $\sim 70$  cm) and to complete a procedure ensuring that stimulus size was relatively stable across participants (see Credit Card procedure in Li et al., 2020, and <https://gitlab.pavlovia.org/Wake/screenscale>).

### DMTS (Session 1, 7, and follow-up)

Stimuli consisted of colored and light grey dots (diameter:  $0.6^\circ$ ), presented on a dark-grey background. Different colors were used for the colored dots in the relevant (i.e., targets) and irrelevant hemifield. In each trial, either two, three, or four colored dots together with grey dots were randomly presented in each hemifield (total number of stimuli presented on the screen:  $24 = 12$  items for each hemifield, comprising colored + grey dots). The dots were positioned inside an invisible  $8$  (rows)  $\times$   $10$  (columns) ( $16^\circ$  width  $\times$   $10^\circ$  height) grid centered at the center of the screen, where a black fixation cross was always present (see also Tagliabue et al., 2022, for further details).

In each trial, above the fixation cross a 100% valid black arrow appeared for 500 ms, pointing randomly and with equal probability towards left or right (i.e., the relevant hemifield). After a 1-second delay, a bilateral memory array appeared for 300 ms, with a subsequent 1,200-ms retention interval. Participants had to memorize the colored dots in the relevant hemifield (targets). A bilateral test array followed: in half of the trials, the relevant hemifield of the test array was identical to that of the memory array (no change condition), while in the remaining half of the trials one target of the relevant hemifield changed color (change condition). Dots in the irrelevant hemifield never changed in color. The test array remained on screen for a maximum of 3 seconds or until response. Participants judged whether the test differed from the memory array by pressing the arrow keys on the keyboard (right for 'same,' left for 'different') with their right hand (Fig. 1b). The inter-stimulus interval was set to 1 second. There were 216

**A****B****C**

**Fig. 1** Training and trial structure. **a** Training timeline for each training group (top: Passive; center: Enumeration; bottom: DMTS). **b** Temporal structure of a DMTS trial with a memory load of three elements in the left

hemifield. **c** Temporal structure of an Enumeration trial with three elements to enumerate. (Color figure online)

total trials (72 trials for each numerosity) divided into 6 blocks of 36 trials each (12 trials per numerosity).

### Adaptive DMTS (Sessions 2 to 6)

Stimulus parameters did not differ from the DMTS outcome task. Each intermediate session was implemented with a

staircase-like procedure: it started with the easiest memory load (i.e., one colored target) and increased in difficulty (i.e., one step up to the next difficulty level by adding one target per block) if the participant's accuracy reached at least 90% at the end of the block. In case the accuracy was below 90%, the participant remained at the same difficulty level (i.e., no change in difficulty level). The training session comprised

10 blocks of 20 trials each. The maximum difficulty that could be reached was a memory load of six elements.

### Adaptive Enumeration (Sessions 2 to 6)

Stimulus size, target colors and spatial arrangement did not differ from the DMTS task. However, no arrow cue was presented before the enumeration array (lasting 300 ms) and colored dots were presented in one hemifield only (i.e., participants did not know in advance the relevant side). The task was to enumerate the colored dots presented. Participants responded, without time limits, through mouse clicking on an answer grid with numbers ranging from 1 to 9 (Fig. 1c). Participants completed a total of 11 blocks of 24 trials each; of these 24 trials, 18 were ‘target numerosities’ of a specific difficulty level (three target numerosities of six trials each), while the remaining trials were ‘nontarget numerosities’. Similar to the adaptive DMTS training, each session started at the easiest difficulty level and advanced to the next level (one step up in the following block) if accuracy was at least 89% with target numerosities only. If accuracy was below 89%, the participant remained at the same difficulty level. There were seven difficulty levels, as defined in Table 1.

### Statistical analyses

A vWM capacity index ( $k$ ) was computed for each memory load, at Session 1, 7 and Follow-up, according to the formula:  $k = [(\text{hit rate} - \text{false alarm rate}) / (1 - \text{false alarm rate})] \times \text{load}$  (Cowan, 2010; Rouder et al., 2011). Load refers to the number of targets to be remembered (i.e., 2, 3, 4), hits were correct ‘different’ responses in ‘change’ trials and false alarms were incorrect ‘different’ responses in ‘no change’ trials.  $K$  values were then averaged across memory loads (Xu et al., 2018); the higher the value, the larger the individual vWM capacity.

Following previous studies with a comparable sample size, participants were divided into low and high performers

(Arciniega et al., 2018; Asseconci et al., 2021; Schiff et al., 2021) according to a median split on the mean  $k$  value computed in Session 1, separately for each group. Notably, low and high performers did not differ in age ( $p = .222$ ) and years of education ( $p = .437$ ).

Improvement (between Session 1 and 7) was investigated through a mixed analysis of variance (ANOVA) on mean  $k$  values, with Training (three levels: Passive, Enumeration, DMTS) and Performance (two levels: Low-Performers, High-Performers) as between-subjects factors and Session (two levels: Session 1, Session 7) as the within-subjects variable. We were not interested in directly comparing improvements between performance groups, but in how training modulated the performance within each performance group (see also Asseconci et al., 2021, for a similar approach). For this reason, we ran training gain comparisons only within each performance group.

On the basis of the results obtained by comparing Session 1 versus Session 7, training effect after three months (follow-up) was investigated only in low-performing individuals. vWM capacity at follow-up was compared with baseline performance via a mixed ANOVA on mean  $k$  values, with Training as between-subjects factor and Session (two levels: Session 1, Follow-Up) as the within-subjects variable.

Additionally, for both Session 1 versus Session 7 and Session 1 versus Follow-Up analyses, we computed training gains (Jaeggi et al., 2011; Jaeggi et al., 2014) to directly compare the improvement between training groups.

Finally, for intermediate training sessions (Sessions 2 to 6) of the Enumeration and Training groups we computed an aggregated performance index that describes an individual’s improvement in adaptive training. The index was obtained by multiplying the difficulty level reached in each block by its relative accuracy, averaged across blocks to obtain one value per session. These values were submitted to a mixed ANOVA with Session (five levels: Sessions 2 to 6) as a within-subjects factor and Performance as a between-subjects variable, separately for each training group.

Significant main and interaction effects were further tested with ANOVAs and Bonferroni–Holm-corrected  $t$  tests ( $p_{\text{bonf-holm}}$ ). In case of significant Mauchly’s or Levene’s tests, corrected Greenhouse–Geisser or Welch  $p$  values are reported, respectively.

## Results

### Session 1 versus Session 7

The ANOVA revealed significant main effects of Performance,  $F(1, 98) = 104.762$ ,  $p < .001$ ,  $\eta_p^2 = .517$ , and Session,  $F(1, 98) = 52.587$ ,  $p < .001$ ,  $\eta_p^2 = .349$ , and significant Session  $\times$  Training,  $F(2, 98) = 5.944$ ,  $p = .004$ ,  $\eta_p^2 =$

**Table 1** Definition of the difficulty levels of the adaptive enumeration task

| Difficulty | Target Numerosities | Nontarget Numerosities |
|------------|---------------------|------------------------|
| 1          | 1 2 3               | 4 5 6 7 8 9            |
| 2          | 2 3 4               | 1 5 6 7 8 9            |
| 3          | 3 4 5               | 1 2 6 7 8 9            |
| 4          | 4 5 6               | 1 2 3 7 8 9            |
| 5          | 5 6 7               | 1 2 3 4 8 9            |
| 6          | 6 7 8               | 1 2 3 4 5 9            |
| 7          | 7 8 9               | 1 2 3 4 5 6            |

*Note.* In each block, 18 trials were “target numerosities” and 6 were “nontarget numerosities”



.108, Session  $\times$  Performance,  $F(1, 98) = 13.732$ ,  $p < .001$ ,  $\eta_p^2 = .123$ , and Session  $\times$  Training  $\times$  Performance,  $F(2, 98) = 3.450$ ,  $p = .036$ ,  $\eta_p^2 = .066$ , interactions.

To investigate the significant Session  $\times$  Training  $\times$  Performance interaction, first follow-up ANOVAs for Session 1 mean  $k$  values were separately conducted in each performance group, to exclude preexisting differences between training groups.

### Low performers

The Training factor was not significant ( $p = .176$ ) in Session 1, suggesting no difference in baseline performance between groups. Given the absence of preexisting differences, follow-up analyses (ANOVAs) were computed on training gains (Session 7 – Session 1 difference). A significant Training effect emerged,  $F(2, 50) = 7.721$ ,  $p_{\text{welch}} = .002$ ,  $\eta_p^2 = .182$ . Post hoc one-sample  $t$  tests against the zero value (i.e., no improvement) indicated that gains were significant for Enumeration,  $t(17) = 4.133$ ,  $p < .001$ ,  $d = .974$ , and DMTS,  $t(16) = 4.330$ ,  $p < .001$ ,  $d = 1.050$ , but not for the Passive group ( $p = .096$ ). Coherently, the pairwise comparisons across groups indicated that Enumeration and DMTS training yielded greater improvement than the Passive group ( $p_{\text{bonf-holm}} = .014$  and  $p_{\text{bonf-holm}} = .006$ , respectively). Training gains between Enumeration and DMTS groups did not significantly differ ( $p_{\text{bonf-holm}} = .509$ ).

### High performers

The effect of Training in Session 1 was significant,  $F(2, 48) = 4.854$ ,  $p = .012$ ,  $\eta_p^2 = .168$ . Subsequent comparisons showed that participants in the Passive group had higher vWM capacity than those in the Enumeration ( $p_{\text{bonf-holm}} = .030$ ) and DMTS ( $p_{\text{bonf-holm}} = .040$ ) groups. No significant difference emerged between Enumeration and DMTS participants ( $p_{\text{bonf-holm}} = .682$ ). To handle this pre-existing difference between groups, we proceeded as follows.

First, we averaged the median  $k$  values of the Enumeration (median = 2.001) and DMTS (median = 2.052) high-performing participants in Session 1 (since these two groups did not differ from each other). Next, we calculated the number of high-performing participants of the Passive group (total  $N = 18$ ) that fell below ( $N = 5$ ) and above ( $N = 13$ ) this averaged median value, highlighting the asymmetrical distribution of the individuals in this group (namely, a symmetrical distribution should be 9–9). Therefore, we randomly removed four out of 13 Passive participants who scored above the averaged median value.<sup>1</sup> We repeated this random sampling 100 times and then proceeded as in the analyses for the low-performing individuals (namely, by conducting a one-way

ANOVA on mean  $k$  values in Session 1 and, subsequently, on training gains, Session 7 – Session 1, with Training as a between-subjects factor). Out of these 100 ANOVAs on mean  $k$  at Session 1, 33 times Training was significant ( $F$  values range: 1.589–4.210;  $p$  values range: .021–.216); however, when checking post-hoc comparisons for these 33 significant Training effects, only three times there was a significant difference between groups (specifically, the Passive group had better performance than the Enumeration group only; significant  $p_{\text{bonf-holm}}$  values range = .048–.050). The 100 follow-up ANOVAs on training gains did not reveal a significant Training effect ( $F$  values range: 1.739–2.074;  $p$  values range: .138–.189). Training gains were also compared against zero (regardless of group) and the 100 one-sample  $t$  tests were always significant ( $t$  values range: 3.214–4.171;  $p$  values range: .0001–.002), indicating a test–retest effect.

In sum, only low-performing participants who trained on the Enumeration and DMTS tasks equally improved their vWM capacity (Fig. 2a). High-performing individuals only showed test–retest effects (Fig. 2b)<sup>2,3</sup>.

### Session 1 versus follow-up

Four participants did not complete the follow-up assessment: three high-performers of the Passive group and one high-performer of the DMTS group.

On the basis of the results obtained from the previous analyses, indicating training effects for low-performers only, training-related changes at follow-up were investigated only in the three low-performing groups.

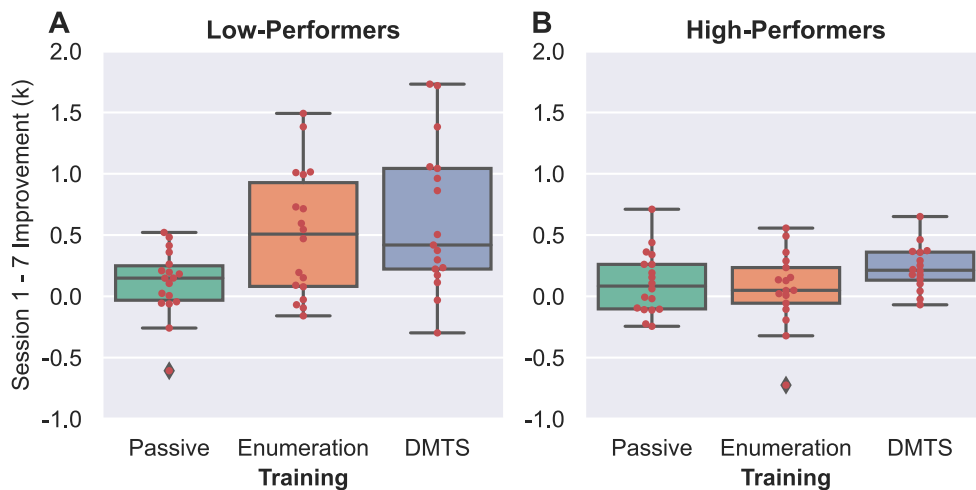
The mixed ANOVA revealed a significant main effect of Session,  $F(1, 50) = 36.680$ ,  $p < .001$ ,  $\eta_p^2 = .423$ , and a significant Session  $\times$  Training,  $F(2, 50) = 3.233$ ,  $p = .048$ ,  $\eta_p^2 = .115$ , interaction.

To investigate the two-way interaction, given the absence of baseline differences in low-performing groups (see Session 1 vs. Session 7 analysis), training gains were computed as follow-up mean  $k$  – Session 1 mean  $k$ . The presence of an actual improvement was tested with a series of one-sample  $t$  tests versus zero. The tests were significant for all groups,

<sup>2</sup> As a control analysis (see Knapp & Schafer, 2009, we additionally conducted a one-way ANCOVA on  $k$  values for Session 7 with Training as between-subjects factor and mean  $k$  values for Session 1 as covariate. As expected, the covariate was significant,  $F(1, 100) = 63.372$ ,  $p < .001$ ,  $\eta_p^2 = .388$ . Importantly, a significant Training effect,  $F(2, 100) = 3.833$ ,  $p = .025$ ,  $\eta_p^2 = .071$ , emerged, thus confirming the presence of training-related changes when controlling for baseline performance.

<sup>3</sup> The inclusion of the Passive control group and the absence of direct comparisons of training gains between the two performance groups should exclude that training effects potentially reflect regression to the mean (Barnett, 2004). As an additional analysis, we correlated the performance in Session 1 with training gains for the Passive group (Ramsay et al., 2018), collapsing across performance level. No significant correlation emerged ( $p = .653$ ), further suggesting that improvements in low-performing Enumeration and DMTS groups are likely due training.

<sup>1</sup> Another approach could have been to test additional participants; this, however, would still have not ensured a homogeneous distribution of the sample.

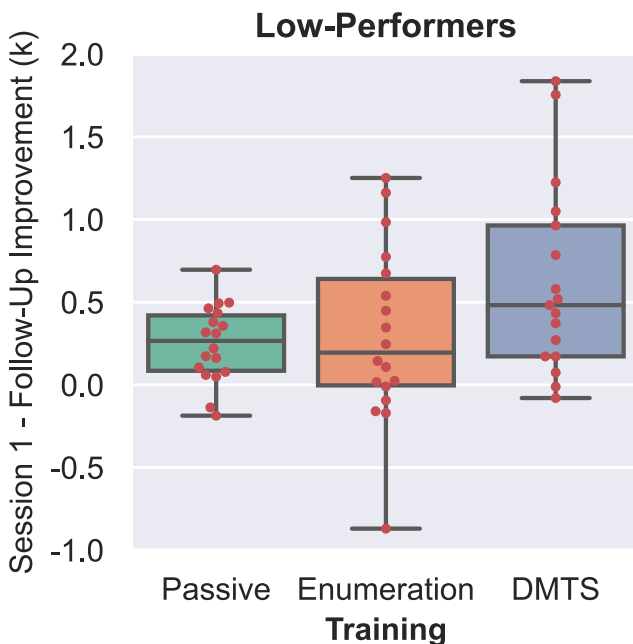


**Fig. 2** Box and whisker plot of the mean vWM improvement (k) between Session 1 and Session 7 in (a) Low-Performing and (b) High-Performing participants. Red dots represent individual subject data, diamonds

indicate data falling outside 1.5 times the interquartile range above the upper and below the lower quartile. (Color figure online)

Passive:  $t(17) = 4.549, p < .001, d = 1.072$ ; Enumeration:  $t(17) = 2.395, p = .028, d = .565$ ; DMTS:  $t(16) = 4.469, p < .001, d = 1.084$ . Subsequent pairwise  $t$ -test comparisons further showed that the DMTS group improved more than the Passive ( $p_{\text{bonf-holm}} = .045$ ) group, but not more than the Enumeration ( $p_{\text{bonf-holm}} = .188$ ) group. Enumeration and Passive groups did not differ ( $p_{\text{bonf-holm}} = .706$ ).

In sum, three months after the end of the experimental sessions, all groups showed a benefit compared with baseline; in addition, there seemed to be a tendency for a training-related larger benefit for the DMTS training group (Fig. 3).



**Fig. 3** Box and whisker plot of the mean vWM improvement (k) between Session 1 and Follow-up in Low-Performing participants. Red dots represent individual subject data. (Color figure online)

## Session 2 to 6

No ceiling effects were evident in the intermediate training sessions of both the Enumeration and DMTS groups.

In the Enumeration group, the mixed ANOVA on the aggregated performance index indicated significant main effects of Session,  $F(4, 128) = 11.895, p < .001, \eta_p^2 = .271$ , and Performance,  $F(1, 32) = 6.537, p = .016, \eta_p^2 = .170$ . The main effect of Performance confirmed that high-performing individuals were better than low performers also during the intermediate adaptive enumeration task. Post hoc polynomial contrasts for the Session factor indicated a significant quadratic trend,  $F(1, 32) = 4.944, p = .033, \eta_p^2 = .134$ , with an increase in performance that reached a plateau at about Session 4.

The mixed ANOVA in the DMTS group revealed significant main effects of Session,  $F(4, 124) = 3.772, p = .006, \eta_p^2 = .108$ , and Performance,  $F(1, 31) = 6.743, p = .013, \eta_p^2 = .183$ , and a significant Session  $\times$  Performance interaction,  $F(4, 124) = 3.049, p = .020, \eta_p^2 = .090$ .

Subsequent ANOVAs in each performance group revealed a significant Session effect in low-performing individuals,  $F(4, 64) = 3.984, p = .006, \eta_p^2 = .199$ , with post hoc polynomial contrasts indicating a significant linear trend,  $F(1, 16) = 11.517, p = .004, \eta_p^2 = .419$ . Also, in high-performing individuals Session was significant,  $F(4, 60) = 2.731, p = .037, \eta_p^2 = .154$ , however no significant trend emerged from polynomial contrasts ( $p = .569$ ). In line with these results, follow-up pairwise comparisons indicated that high-performing individuals outperformed low-performers during the first two training sessions ( $p_{\text{bonf-holm}} = .024$  and  $p_{\text{bonf-holm}} = .015$ , respectively), but they did not differ in the last three sessions ( $p_{\text{bonf-holm}} > .081$ ).

In sum, during the Enumeration training participants improved their performance across sessions until reaching a

plateau, regardless of baseline performance (Fig. 4a). In the DMTS training, only low-performing older adults exhibited a linear performance enhancement (Fig. 4b).

## Discussion

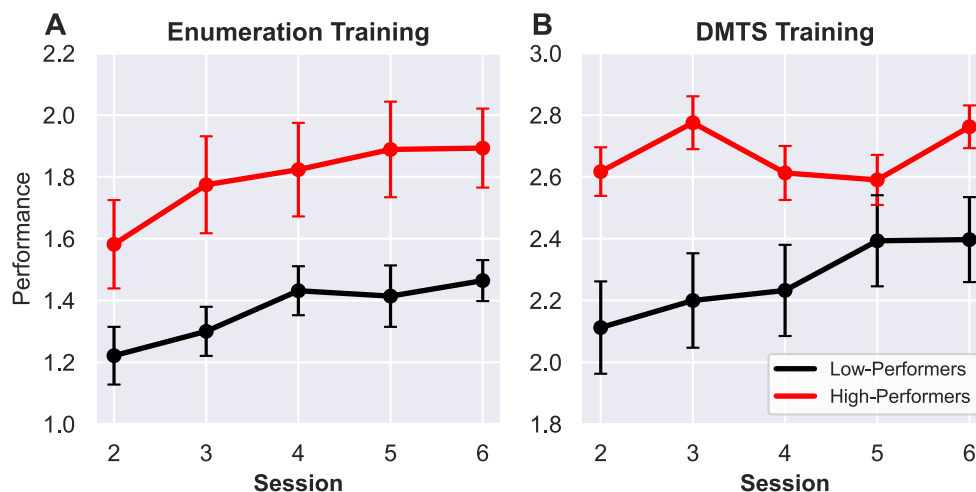
The main goal of the present study was to investigate whether a short online training primarily taxing attention would be effective to improve vWM of older individuals.

The results showed that, after the end of the intervention (i.e., in Session 7), the Enumeration training was able to induce an increase in the vWM capacity of low-performing older participants. Notably, this observed enhancement in vWM was comparable to what was attained by participants who practiced the DMTS task across the intermediate sessions (i.e., practice effect). These results seem to confirm an existing link between attention processes (here, with specific reference to attentive individuation of target items in the enumeration task) and vWM capacity (Rutman et al., 2010; Vogel et al., 2005; Zanto & Gazzaley, 2009). Previous training studies of other attentional aspects (mainly distractor filtering and in younger individuals) produced mixed results concerning benefits to vWM, from negative evidence when using multiple-object tracking (MOT; Arend & Zimmer, 2012; Thompson et al., 2013) to positive evidence when using DMTS where targets and distracters were intermixed (Shin et al., 2015), or with search tasks based on feature integration (Schmicker et al., 2016, and sham tDCS stimulation group in Schmicker et al., 2021).

Despite enumeration in the subitizing range primarily relies on attentional mechanisms (see Chen et al., 2022, for a recent meta-analysis), to-be-enumerated items likely enter the vWM buffer for mapping the items to a symbolic quantity value. Indeed, previous studies showed that the EEG marker

of item maintenance in the vWM buffer (i.e., the CDA) is elicited also during enumeration (e.g., Pagano et al., 2014; Pagano & Mazza, 2012; Poncet et al., 2016). Specifically, EEG markers of perceptual, attentive and WM stages are involved in enumeration tasks; however, it is the attentive stage (as reflected in the N2pc) that correlates with the estimated individual inflection point that determines the individual subitizing limit (Pagano et al., 2014). This result suggests that item individuation is the primary mechanism in rapid enumeration and in the generation of the subitizing phenomenon. Additionally, the temporal sequence of the occurrence of the N2pc and the CDA indicates that vWM capacity limits are not the ultimate cause of subitizing, since a capacity-limit-related modulation is already apparent at the earlier, individuation stage. Overall, the accumulating findings indicate that attentive individuation is the key component in many tasks involving multiple objects (e.g., enumeration, DMTS evaluation, multiple object tracking; Drew & Vogel, 2008; Ester et al., 2012; Störmer et al., 2013). Specifically, once the elements are individuated, they are made available to further cognitive operations likely involving WM for different purposes (e.g., mapping the individuated set of elements onto a symbolic value for enumeration; comparing the individuated elements with a new set of objects to detect a change for DMTS).

In line with this view, the current data seem to complement previous EEG findings (Tagliabue et al., 2020), where practice of the same DMTS task led to a vWM improvement and a concurrent neural modulation of ERP components reflecting attentive individuation (i.e., the N2pc; Eimer, 1996) and active maintenance of items in the memory buffer (i.e., the CDA; Vogel & Machizawa, 2004). Moreover, there is evidence of suboptimal functioning of the object individuation mechanism in aging (Lorenzo-López et al., 2008; Pagano et al., 2015; Störmer et al., 2013), and Störmer and co-authors (2013)



**Fig. 4** Performance at each intermediate session (= average of [difficulty \* accuracy at each block]) in (a) the Enumeration and (b) DMTS groups, for both Low- and High-Performers. Vertical bars represent standard error. (Color figure online)



suggested that its reduced functionality can be responsible for the lower vWM capacity exhibited by older participants in DMTS tasks. Taken together, these data thus seem to suggest that becoming more proficient in item individuation aids the subsequent memorization phase in a DMTS task. We can speculate that enhanced attentive individuation allows participants to retain either more elements or a better representation of them (i.e., with more fine-grained resolution).

Follow-up results (namely, after 3 months) need to be interpreted with caution. Indeed, post-hoc comparisons indicated a significant difference in training gains between the Passive and the DMTS groups, but the Enumeration group did not differ from either of them. Thus, while 3 months after intervention completion there seems to be a tendency in low-performing older adults for benefit after the DMTS training, future research will establish the reliability of this trend. Indeed, training with a different task might induce cognitive improvements that are more fragile and susceptible to deterioration over time, and further investigations could address this claim. In addition, the training regimen proposed here was brief (i.e., five sessions), leaving open the possibility that increasing the number of training sessions could produce more solid and long-lasting effects with respect to the present ones. Indeed, training effects can be dose-dependent (Ophey et al., 2020) and there is evidence of longer training duration having advantages on effects both measured in different tasks from those used as training and at follow-up (Brum et al., 2020).

A final issue concerns the influence of individual differences on training outcomes. As we hypothesized, low-performing older adults at baseline were those who benefited more from the training intervention (both in the Enumeration and the DMTS group), while high performers only showed test–retest effects. These results align with previous literature (e.g., Borella et al., 2017; Traut et al., 2021; Zinke et al., 2014) indicating that process-based WM training procedures (as the current one) are more beneficial for individuals starting from lower cognitive level. Training allows them to more efficiently master cognitive skills that were not fully mastered before the intervention, as predicted by the compensation view. On the other hand, high-performers might have already mastered such cognitive skills to a satisfying degree, thus leaving little room for improvement (Lövdén et al., 2012).

Some limitations of the present study must be listed. First, we used the same stimulus material (i.e., colored dots) across pretraining, posttraining and training tasks, which might resemble perceptual training regimens (e.g., Berry et al., 2010; Li et al., 2017; Mishra et al., 2015) that are based on the repetition of exactly the same stimuli and produce highly specific perceptual improvements (e.g., in contrast or orientation discrimination; Maniglia & Seitz, 2018). However, if effects were mainly due to increased stimulus familiarity (as can be expected from perceptual training), then larger improvements should have been evident also in high-performers of the two

training groups (e.g., Li et al., 2017) with respect to the Passive group, which was not the case in the present study. Moreover, even if follow-up data need to be taken with caution, DMTS and Enumeration training-related effects slightly differed, despite the same amount of exposure to stimuli, which should not occur if the benefit were due exclusively to increased stimulus familiarity. The second limitation of the study is that we did not include as an additional outcome measure another vWM task. This would have allowed us to test whether training such attention-based mechanism produces positive benefits that generalize across vWM tasks.

As a concluding remark, the feasibility of this type of training intervention is worth to be discussed. Indeed, the whole training procedure occurred online, without direct contact between participants and experimenters. Despite the ‘no-contact,’ fully online nature of the intervention, the majority of older participants could easily perform all the training sessions, with an acceptable dropout rate of 17%. Such results are promising and encourage the implementation of more online cognitive training programs addressed to older adults, especially in the light of the current pandemic situation.

## Conclusion

We showed that, after a short cognitive intervention, vWM capacity can improve in older adults when attentive individuation is primarily trained, thus confirming its contribution to vWM functioning. However, this beneficial effect was not clearly present at follow-up, which might suggest that a larger training dose is needed to obtain more stable cognitive changes over time. Finally, our results add to the bulk of evidence indicating that baseline interindividual differences represent a crucial variable in determining training outcomes and that such variability needs to be fully accounted for when implementing effective training interventions.

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**Author's contribution** C.F.T. and V.M. designed the study. C.F.T. programmed the tasks. G.V. collected the data. C.F.T. and G.V. performed the data analysis and interpretation under the supervision of V.M. C.F.T. and V.M. drafted the manuscript, and all authors approved the final version of the manuscript for submission.

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## Declarations

**Competing interests** The authors declare that they have no competing interests.

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